

Nucleotide-resolution prediction of ribosome occupancy from transcriptome sequences across cellular contexts

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Abstract

Translation links RNA sequence to protein output, but existing models typically address translation efficiency, ribosome dynamics or ORF discovery as separate tasks. Here we present TRACE (Translation Resolution Across Cell Environments), which predicts a nucleotide-resolution ribosome occupancy profile from full-length RNA sequence and cellular context and is annotation-free at inference. TRACE was trained on 1.8 million transcript–context profiles from 73 cellular contexts across three species. Across held-out transcripts, the same output supports translation dynamics prediction, CDS-level translation efficiency estimation and candidate ORF identification, with performance evaluated against task-specific models and experimental baselines. Prediction and attention patterns were associated with regulatory features across transcript architecture, initiation motifs and codon positions. Because CDS and reading-frame supervision were used during training, these associations are not interpreted as de novo mechanistic discovery. The unified representation also supports forward design of CDS variants in cultured cells and reverse discovery of translated ORFs from tumor transcriptomes without matched tumor Ribo-seq or mass spectrometry for candidate nomination. TRACE thus connects prediction, design and discovery through one translation representation.

Introduction

Protein output cannot be inferred reliably from transcript abundance alone because translation varies across transcripts and cellular contexts. Ribosome profiling has revealed translated regions in canonical coding transcripts and in many RNAs previously classified as noncoding. These observations expand the biological scope of translation and motivate predictive models that

27 resolve where ribosomes occupy each transcript. Such models could support both sequence
28 design and the discovery of translated open reading frames (ORFs).

29 Ribosome profiling measures ribosome occupancy at nucleotide resolution, but its experimental
30 requirements limit coverage across tissues, cell states and clinical samples. Existing
31 computational methods address parts of this problem. Some estimate transcript-level translation
32 efficiency or local CDS density, whereas others identify ORFs from Ribo-seq or predict
33 translation boundaries from sequence. However, these capabilities are usually separated across
34 models, and several approaches require CDS annotations, fixed sequence windows or
35 experimental Ribo-seq inputs. A unified model must predict a full-length occupancy profile
36 while remaining annotation-free at inference.

37 This task is challenging for three reasons. First, translation depends on sequence features ranging
38 from local initiation motifs to full-transcript architecture. Second, the same RNA can be
39 translated differently across cellular contexts. Third, training data from different Ribo-seq studies
40 vary in coverage, protocol and P-site assignment. Addressing these challenges requires a
41 harmonized dataset, a variable-length sequence model and explicit cellular context conditioning.
42 It also requires evaluation on held-out transcripts that do not overlap the training set.

43 Here we present TRACE (Translation Resolution Across Cell Environments), a transformer that
44 predicts a nucleotide-resolution ribosome occupancy profile from full-length RNA sequence and
45 cellular context. TRACE was trained on 1.8 million transcript–context profiles from 73 cellular
46 contexts across three species. Among the methods compared in Supplementary Table 3, only
47 TRACE combines full-length variable-length input, cellular context conditioning, unified multi-
48 species modeling, nucleotide-resolution occupancy prediction and annotation-free inference.
49 This translation representation supports tasks that are commonly separated. Within a known
50 CDS, occupancy magnitude and position provide an objective for forward sequence design.
51 Across an unannotated transcript, the same profile supports reverse discovery by localizing
52 candidate translated ORFs. We evaluate these capabilities across translation dynamics,
53 translation efficiency, ORF identification, cultured-cell mRNA design and tumor neoantigen
54 candidate prioritization.

55 Results

56 A nucleotide-resolution dataset of ribosome occupancy across cellular contexts

57 The construction of a reliable training dataset proved to be a non-trivial component of this work.
58 Raw Ribo-seq data present several challenges for quantitative modeling. First, read coverage per
59 transcript is typically sparse, with many nucleotide positions receiving zero counts, particularly
60 in transcripts with low ribosome occupancy. Second, experimental protocols differ across
61 laboratories in nuclease choice, size selection, and library preparation, introducing systematic
62 biases in the distribution of ribosome-protected fragment lengths. Third, Ribo-seq signal
63 conflates translational activity with transcript abundance: a highly expressed transcript with
64 modest translation efficiency can produce more Ribo-seq reads than a lowly expressed transcript
65 that is efficiently translated. Disentangling these factors required careful preprocessing.

66 We aggregated public Ribo-seq and matched RNA-seq datasets covering human, rhesus
67 macaque, and mouse. After curation, the collection comprised 1,355 Ribo-seq runs and 475
68 RNA-seq runs from 73 cellular contexts, together with structured metadata recording species,
69 tissue of origin, and experimental conditions.

70 All data passed through a single processing workflow designed to produce comparable
71 nucleotide-resolution ribosome occupancy profiles across experiments. The first stage performed
72 quality control: runs with low CDS read enrichment were discarded, and reads from the same
73 cell type were merged to maximize per-transcript coverage. In the second stage, transcripts with
74 mean depth below 0.1 or coverage below 0.1 were excluded, retaining only those supported by
75 splice-junction-spanning reads. The third stage addressed P-site assignment, a critical step
76 because the offset between the ribosome-protected fragment end and the actual P-site varies with
77 fragment length and experimental protocol. We trained a LightGBM classifier on features
78 derived from read length, surrounding sequence context, and annotated CDS frame preference to
79 predict the P-site position for each fragment. Datasets with a three-nucleotide periodicity score
80 below 0.5 on annotated CDS regions were removed, ensuring that the retained profiles reflected
81 genuine ribosome translocation patterns.

82 To isolate translational signal from RNA abundance, we applied compositional regression to
83 estimate a translation efficiency scaling factor for each transcript. Multiplying normalized P-site
84 profiles by this factor yielded nucleotide-resolution ribosome occupancy profiles after

accounting for expression level. The dataset comprises 1.8 million transcript–context profiles. A transcript contributes one profile for each cellular context with an available ribosome-density profile. Canonical protein-coding transcripts showed dense, periodic occupancy within annotated CDS regions.

Following transcriptional normalization, variation in translation efficiency across transcripts exceeded variation among cellular contexts. Between-species differences also exceeded within-species differences among cellular contexts. These patterns are consistent with a substantial contribution of cis sequence to translational output. They do not establish causal dominance because species differ in many sequence-independent factors. Related cellular contexts nevertheless clustered by translation efficiency across species, supporting an additional contribution from trans-acting cellular programs. We therefore used RNA sequence as the model backbone and cellular context as a conditioning signal for context-specific modulation (Fig. 1).

Nucleotide-resolution prediction of ribosome occupancy from transcriptome sequences

Motivated by the substantial contribution of cis sequence suggested by the cross-species analysis, we designed the TRACE backbone to process full-length RNA directly. Bidirectional multi-head self-attention with Rotary Position Embedding captures sequence associations across local motifs and long transcript regions. Flash Attention supports memory-efficient processing of variable-length transcripts. Adaptive Layer Normalization modules introduce cellular context conditioning throughout the transformer stack. This architecture uses RNA sequence as a shared representation and cellular context to modulate nucleotide-resolution predictions (Fig. 2a).

TRACE accepts three inputs: full-length RNA sequence, a species identifier and a cellular context vector derived from gene expression. The RNA contains the 5' UTR, CDS and 3' UTR and is tokenized as individual nucleotides. Rotary position embeddings encode 5'-to-3' position without a fixed-length matrix. Flash Attention and length-aware batching reduce memory and padding costs for transcripts ranging from hundreds to tens of thousands of nucleotides. The output has the same length as the RNA and assigns each position an expected normalized ribosome occupancy (Fig. 2a).

Cellular context was incorporated through Adaptive Layer Normalization modules rather than separate models for each cell type. A low-dimensional embedding of gene expression modulates

the scale and shift parameters at each transformer block. The sequence-processing parameters are shared across contexts, whereas conditioning adjusts the predicted occupancy profile for the supplied cellular context.

Training used position-level density, a CDS frame-aware TE objective and a transcript-level ranking objective. Annotated CDS boundaries and reading frames were used to construct the training losses, but were not supplied during inference. Codon identity was represented only through the nucleotide sequence, and RNA-binding-protein motif labels were not provided to the sequence-only model. TRACE is therefore annotation-free at inference: it receives RNA sequence, species and cellular context. Annotations were used only downstream when a benchmark required CDS-level aggregation.

We evaluated TRACE on held-out transcripts from chromosomes 20, 21, 22 and Y. These transcripts and their associated context profiles were excluded from training, model selection and early stopping. During inference, neither CDS nor reading-frame annotation was provided. Predicted profiles showed three-nucleotide periodicity and occupancy concentrated in annotated CDS regions (Fig. 2b,c). Because CDS and frame labels contributed to training, these results indicate generalization of supervised translation patterns to held-out transcripts rather than unsupervised discovery of frame structure.

Annotation-free at inference, TRACE assigned different periodicity scores to transcript classes. Housekeeping transcripts had mean scores above 0.5, whereas long noncoding RNAs had scores below 0.5 (Fig. 2d). This separation is consistent with expected biotype differences, although the training objectives included CDS and reading-frame supervision.

For the housekeeping transcript ENST00000332859, the predicted and observed profiles had a Spearman ρ of 0.80 across nucleotide positions. TRACE predicted little occupancy for the noncoding RNA ENST000000654422. For ENST000000789734, it predicted a localized occupancy region that overlapped an experimentally observed Ribo-seq signal. This example is consistent with coding potential in a transcript annotated as noncoding, but it does not alone establish stable protein production.

TRACE uses one nucleotide-resolution ribosome occupancy profile as a shared translation representation. For a known CDS, the profile can be aggregated to estimate translation efficiency and can score sequence variants for forward design. For an unannotated transcript, the same profile can be scanned for periodic occupancy to support reverse discovery of candidate ORFs.

This design connects prediction, design and discovery without separate task-specific representations.

Benchmarking a unified translation representation across tasks

We evaluated the shared TRACE output on translation dynamics, translation efficiency and ORF identification (Fig. 3a). These tasks are commonly addressed by separate models with different inputs and output resolutions. Within the prior-work comparison in Supplementary Table 3, TRACE is the only evaluated model combining nucleotide-resolution occupancy prediction, explicit cellular context conditioning, unified multi-species modeling and annotation-free inference. We compared TRACE with task-specific published models and with empirical baselines derived from replicate and cross-study measurements.

For each transcript, we computed Spearman ρ between predicted and observed P-site profiles across nucleotide positions. Reference baselines compared Ribo-seq replicates within a study and data from the same cell type across studies. In the evaluated datasets, TRACE approached the cross-study baseline across transcript lengths and ribosome loads (Fig. 3b,c). Its correlations were higher than those of the retrained sequence-only Riboformer and RiboMIMO baselines. Performance was also consistent across evaluated cellular contexts, including SW480, which was excluded from training (Fig. 3d).

We next evaluated CDS-level translation efficiency using three orthogonal measurements. TRACE predictions showed among the highest Spearman ρ values in comparisons with polysome-derived ribosome load (Fig. 3e). They also ranked in the top group for protein-to-RNA ratios and for SILAC-measured protein synthesis in an independent mouse fibroblast dataset (Fig. 3f,g). These analyses assess transcript-level output derived from the same nucleotide-resolution profile.

We evaluated ORF identification against public mass spectrometry evidence. Across the tested tissues, TRACE produced higher F1 scores than the evaluated Ribo-seq-based callers and sequence-derived baselines. Unlike the Ribo-seq-based tools, TRACE did not require a sample-specific ribosome-profiling input for candidate prediction. The comparison supports the use of nucleotide-resolution predictions for discovery when experimental translation data are unavailable.

Public benchmarks can be confounded by differences in sample origin, protocol and processing. We therefore generated a matched human fetal-brain dataset containing Iso-Seq, Ribo-seq and

177 tandem mass spectrometry. On full-length isoforms, TRACE predictions correlated with
178 experimental Ribo-seq across translation-abundance ranges, whereas Translatomer reached a
179 Pearson r of approximately 0.1. The precision of top-ranked short-ORF predictions was
180 comparable to RiboTIE when evaluated against mass spectrometry evidence. Together, these
181 measurements provide matched support for the sequence-to-occupancy-to-peptide analysis chain
182 without implying complete proteome recovery.

183 **Translation-associated patterns across multiple sequence scales**

184 We examined whether TRACE predictions reflect translation-associated patterns across
185 transcript architecture, initiation regions and codon positions. CDS boundaries and reading
186 frames contributed to training losses, so these analyses do not test whether the model discovered
187 translation rules without supervision. We treat prediction sensitivity and attention weights as
188 associative descriptions of model behavior, not as causal or mechanistic explanations.

189 TRACE-predicted translation efficiency was negatively associated with 5' UTR, CDS and 3'
190 UTR length (Fig. 4a). Predicted efficiency also decreased as RNA minimum free energy became
191 lower, a pattern consistent with reduced translation of more stable RNA structures (Fig. 4b).
192 These features were not supplied as scalar model inputs, but the observed associations do not
193 establish causality.

194 Aggregated attention weights were enriched near the annotated start region, across the CDS and
195 near the stop region (Fig. 4c). Attention near the start codon was associated with positions in the
196 Kozak consensus, and stronger Kozak contexts coincided with higher predicted initiation (Fig.
197 4d). Because attention is not a causal attribution method, these patterns are interpreted as
198 associations between model weighting and known initiation features.

199 Frame-resolved attention differed across CDS positions, with higher weights at frame 0 and
200 frame 1 than at frame 2 (Fig. 4e). This pattern should be interpreted in light of the frame-aware
201 supervision used during training. In silico synonymous substitutions at third codon positions
202 produced smaller predicted changes than substitutions at first or second positions (Fig. 4f). The
203 result describes model sensitivity to codon position and does not by itself establish an elongation
204 mechanism.

205 In silico removal of upstream AUG codons increased predicted occupancy over the downstream
206 CDS (Fig. 4g). The predicted change was larger for out-of-frame than in-frame upstream ORFs.
207 At matched positions, AUG in a strong Kozak context had a larger predicted effect than CUG or

208 GUG (Fig. 4h). These perturbation responses are consistent with established initiation-associated
209 patterns but remain model-based associations.

210 **Forward design of CDS translation in specific cellular contexts**

211 The shared occupancy representation also provides an objective for forward sequence design.
212 Within a known CDS, the predicted signal can score sequence variants before synthesis and can
213 target different translation levels in a specified cellular context.

214 We designed ten firefly luciferase and mCherry mRNAs with graded predicted translation levels.
215 After transfection into HEK293T cells, measured protein output correlated with predicted
216 translation intensity (Pearson $r > 0.9$ for both reporters). This result supports quantitative
217 sequence design in cultured cells rather than only binary high-versus-low classification.

218 We next optimized the Gaussia luciferase coding sequence for expression in HEK293T cells.
219 The TRACE-designed mRNA produced higher measured expression than the wild-type,
220 RiboDecode-designed and LinearDesign-designed sequences in the evaluated experiment. This
221 comparison tests whether position-level CDS predictions can guide forward design beyond
222 codon-usage or structure-only objectives.

223 We also tested design specific to each cellular context using HEK293T, HeLa and HepG2 cells.
224 Each reporter sequence was designed for higher predicted expression in one target context while
225 retaining the same encoded protein. The measured expression patterns followed the intended
226 context preference, supporting sequence design conditioned on cellular context in cultured cells.

227 **Reverse discovery of non-canonical tumor neoantigen candidates from** 228 **transcriptomes**

229 Tumor-specific peptides can arise from non-canonical ORFs, retained introns, alternative splice
230 products and endogenous retroelements. These candidates extend beyond somatic mutation-
231 derived epitopes, but their prioritization requires evidence that the source RNA may be
232 translated. This setting tests reverse discovery from transcripts whose coding regions are not
233 known in advance.

234 Where forward design evaluates variants within a known CDS, reverse discovery scans an
235 annotation-free occupancy profile to localize candidate translated ORFs. We applied this
236 workflow to paired tumor and adjacent-normal RNA-seq from two patients. Transcript assembly,
237 tumor-specificity filtering, TRACE ORF prediction and HLA class I binding prediction yielded

ten candidate neoepitopes. Candidate nomination required neither matched tumor Ribo-seq nor mass spectrometry.

Patient peripheral blood mononuclear cells were stimulated with candidate peptides and evaluated by IFN- γ ELISpot. Reactive candidates were further examined in patient-derived tumor-organoid co-cultures. These assays provide functional support for selected candidates without matched tumor Ribo-seq or mass spectrometry. They do not directly quantify endogenous peptide abundance or replace immunopeptidomic confirmation.

Discussion

This study provides 1.8 million transcript–context profiles from 73 cellular contexts across three species. A uniform processing pipeline harmonized heterogeneous Ribo-seq studies into comparable nucleotide-resolution ribosome occupancy profiles. The resulting resource supports model training and evaluation across transcripts and cellular contexts, although coverage remains uneven across tissues.

TRACE uses full-length RNA sequence as a shared backbone and cellular context as a conditioning signal. Cross-species and cross-context patterns are consistent with a substantial cis-sequence contribution, but they do not establish that sequence causally dominates translation. Likewise, CDS and reading-frame supervision contributed to training. Periodicity, CDS enrichment and attention patterns should therefore be interpreted as generalization and association rather than de novo mechanistic discovery.

TRACE predicts ribosome occupancy rather than downstream protein stability, folding, modification or degradation. Prediction accuracy remains limited for transcripts with very low ribosome density and for cellular contexts underrepresented in the training data. The model does not explicitly represent RNA modifications, measured RNA structure, RNA-binding-protein occupancy or subcellular localization. For neoantigen candidates, functional assays do not replace direct evidence of endogenous peptide presentation.

A single translation representation connects the two application directions. Within annotated CDSs, position-level occupancy supports forward design of sequences with specified translation levels and preferences among cellular contexts. Across unannotated transcripts, the same nucleotide-resolution output supports reverse discovery of candidate ORFs and neoantigens. This

267 shared logic links prediction, design and discovery while preserving the evidence boundaries of
268 each application.

269 Future work could incorporate RNA structure, modification state, RNA-binding-protein
270 occupancy and subcellular localization. Generative models may extend forward design beyond
271 the sequence variants evaluated here. Direct immunopeptidomic measurements could strengthen
272 reverse-discovery validation. Pretrained RNA representations may also improve generalization to
273 species and cellular contexts that are sparsely represented in the current dataset.

274 **Methods**

275 **Matched Ribo-seq and RNA-seq data collection and curation**

276 We collected 1,236 human Ribo-seq datasets, 14 rhesus macaque Ribo-seq datasets and 9 mouse
277 Ribo-seq datasets, together with 704 matched RNA-seq data. All input data were obtained as raw
278 FASTQ files. For Ribo-seq, adapters were first detected and removed with fastp (version 0.23.4).
279 When automatic detection was unsuitable, adapter sequences were obtained from the source
280 publication and removed with cutadapt (version 4.5) or Trimmomatic (version 0.39). To reduce
281 rRNA and tRNA contamination, trimmed reads were aligned with Bowtie2 (version 2.5.2) to
282 species-specific rRNA and tRNA reference sequences (human, hg38, Ensembl V114; rhesus
283 macaque, Ensembl V101, rheMac10; mouse, mm39, Ensembl V109). Unmapped reads were
284 retained and aligned to the corresponding reference genome with STAR (version 2.7.9a). We
285 retained Ribo-seq datasets for which more than 60% of aligned reads mapped within annotated
286 coding sequences (CDSs).

287 Matched RNA-seq reads were adapter-trimmed with fastp (version 0.23.4) and aligned to the
288 corresponding reference genome with STAR (version 2.7.9a). Library strandedness was inferred
289 from the aligned BAM files using RSeQC (version 5.0.1). Transcript-level read counts were then
290 quantified with featureCounts (version 2.0.6) using the same species-specific gene annotation as
291 for genome alignment.

292 Metadata for each Ribo-seq dataset and its matched RNA-seq dataset, including species,
293 biological source, nuclease and accession, are provided in Supplementary Table 1.

Unified Ribo-seq processing and normalization

Ribo-seq BAM files that passed dataset-level quality control were grouped by tissue or cell line and merged with samtools (version 1.19.2). The resulting dataset comprised 73 cellular contexts, including 29 tissues (22 human, 3 mouse and 4 rhesus macaque) and 44 cell lines with matched RNA-seq data.

For each cellular context, alignment of ribosome protected fragment (RPF) support was used to select a parsimonious set of representative transcripts that captured supported exons and splice junctions, while retaining MANE Select isoforms. Transcripts that did not explain additional informative alignment features were removed, reducing isoform redundancy before profile construction.

We then used a nuclease-aware LightGBM classifier to infer a read-specific P-site offset from fragment length and sequence context at both RPF ends. This formulation accommodated libraries generated under different nuclease conditions, including RNase I and micrococcal nuclease (MNase), without imposing a single length-specific offset rule. P-site assignments were aggregated at nucleotide resolution.

To construct the datasets, informative transcript–context profiles were retained using matched RNA-seq abundance and P-site depth and coverage; frame periodicity was additionally required only for transcripts with a valid annotated CDS, whereas full-length criteria were used otherwise. Besides, non-zero P-site counts were processed using the winsorization method at the 99.5th percentile to minimize the impact of PCR bias. Within each cellular context, transcript-level RPF and matched RNA-seq counts were augmented by a pseudocount of 10 and centered-log-ratio (CLR) transformed. An ordinary least-squares regression of CLR-transformed RPF counts on CLR-transformed RNA-seq counts yielded residual r_t , and $\exp(r_t)$ provided a multiplicative transcript-level scale for translation not explained by measured RNA abundance. For transcript t and nucleotide position l , the RNA-abundance-adjusted relative ribosome occupancy target was defined as:

$$y_{tl} = \log \left[1 + \frac{c_{tl}}{\hat{c}_t} \exp(r_t) \right]$$

$$\hat{c}_t = \frac{1}{|P_t|} \sum_{l \in P_t} c_{tl}, r_t = \text{clr}(R_t + 10) - [\beta_0 + \beta_1 \text{clr}(A_t + 10)].$$

322 Here c_{tl} denotes the winsorized P-site count, P_t is the set of positions with a positive winsorized
323 count, $\hat{c}_t$ is the mean over those positions, R_t is the transcript-level RPF count and A_t is the
324 matched RNA-seq count. The regression coefficients β_0 and β_1 were estimated separately within
325 each cellular context. Thus, y_{tl} is a dimensionless signal that combines within-transcript
326 positional occupancy with an RNA-abundance-adjusted transcript-level scale; it is not an
327 absolute ribosome density or a site-occupancy probability.

328 This workflow yielded 1.8 million transcript–context profiles from 73 cellular contexts across
329 human, rhesus macaque and mouse. A transcript observed in multiple cellular contexts
330 contributed one profile per context. Detailed transcript selection, P-site inference, profile filtering
331 and normalization parameters are provided in Supplementary Methods.

332 **TRACE model architecture**

333 TRACE predicts a nucleotide-resolution ribosome occupancy profile from full-length RNA
334 sequence, a species label and a 16,840-gene expression vector. Nucleotides were one-hot
335 encoded in A, C, G and T (U) channels, with ambiguous bases represented by all-zero vectors. A
336 linear layer projected each nucleotide to 384 dimensions, followed by layer normalization,
337 GELU activation and dropout ($p = 0.1$). The 128-dimensional species embedding was
338 concatenated with the expression vector and passed through a 64-dimensional bottleneck. This
339 bottleneck comprised a linear layer, layer normalization and GELU activation, followed by a
340 linear projection to a 16-dimensional conditioning vector.

341 Token embeddings passed through 12 bidirectional transformer encoder layers. Each layer used
342 16-head self-attention (head dimension, 24) and a position-wise feed-forward network (hidden
343 dimension, 768). Rotary Position Embedding used a base frequency of 10,000, an NTK-aware
344 scaling factor of 8 and a maximum cached context of 30,000 nucleotides. AdaLN-Zero converted
345 the 16-dimensional conditioning vector into scale, shift and residual-gating parameters for the
346 attention and feed-forward sublayers. Flash Attention removed padded positions before attention
347 and restored them afterwards, with scaled dot-product attention used as a fallback. The
348 prediction head comprised layer normalization, a 384-dimensional linear layer, GELU, dropout
349 ($p = 0.1$), a scalar output layer and ReLU activation.

350 TRACE model training

351 Pretraining comprised 1.8 million transcript–context profiles from 73 cellular contexts across
 352 human, rhesus macaque and mouse. Each transcript–context observation counted as one profile;
 353 a transcript measured in ten cellular contexts therefore contributed ten profiles. Chromosome-
 354 disjoint partitions were defined separately for each species. For human data, chromosomes 1–17
 355 defined the training set, chromosomes 18, 19 and X the validation set, and chromosomes 20, 21,
 356 22 and Y the test set. For rhesus macaque data, chromosomes 1–17 and X defined the training
 357 set, chromosomes 18 and 19 the validation set, and chromosomes 20 and Y the test set. For
 358 mouse data, chromosomes 1–15 and X defined the training set, chromosomes 16 and 17 the
 359 validation set, and chromosomes 18, 19 and Y the test set. All profiles from the same transcript
 360 remained in one partition. Test partitions were used only for final evaluation.

361 The training objective combined a token-level profile loss, a frame-aware translation-efficiency
 362 (TE) loss and a transcript-ranking loss. For transcript i , V_i denotes its valid nucleotide positions;
 363 l indexes nucleotide positions within the transcript; and p_{il} and t_{il} denote predicted and observed
 364 occupancy, respectively. The token-level loss was the mean Smooth L1 loss over valid positions:

$$365 \quad L_{micro}^{(i)} = \frac{1}{|V_i|} \sum_{l \in V_i} \text{SmoothL1}(p_{il}, t_{il}; \beta_{Huber} = 1)$$

366 For transcript i , the loss region R_i was the annotated CDS when available and all valid transcript
 367 positions V_i otherwise. The frame origin a_i was the first nucleotide of R_i . Thus, the first
 368 nucleotide of an unannotated transcript was assigned to frame 0, with subsequent nucleotides
 369 cycling through frames 1, 2 and 0. Positions assigned to frame $f \in \{0, 1, 2\}$ were:

$$370 \quad F_{if} = \{l \in R_i : (l - a_i) \bmod 3 = f\}$$

$$371 \quad \hat{p}_{if} = \frac{1}{|F_{if}|} \sum_{l \in F_{if}} p_{il}, \hat{t}_{if} = \frac{1}{|F_{if}|} \sum_{l \in F_{if}} t_{il}$$

$$372 \quad L_{TE}^{(i)} = \frac{1}{3} \sum_{f=0}^2 (\hat{p}_{if} - \hat{t}_{if})^2$$

373 The ranking loss was computed within each mini-batch by evaluating every ordered pair (i, j) of
 374 transcript–context profiles. Each comparison used the mean signal over its loss region, R_i or R_j .
 375 This was the CDS mean for annotated transcripts and the full-length mean over valid positions

otherwise. A pair contributed to the loss only when its observed means differed by more than $\delta=0.005$:

$$\dot{p}_i^R = \frac{1}{|R_i|} \sum_{l \in R_i} p_{il}, \dot{t}_i^R = \frac{1}{|R_i|} \sum_{l \in R_i} t_{il}$$

$$P = \{(i, j) : \dot{t}_i^R - \dot{t}_j^R > \delta\}, \delta = 0.005$$

$$L_{rank} = \frac{1}{|P|} \sum_{(i, j) \in P} \max(0, \delta - (\dot{p}_i^R - \dot{p}_j^R))$$

If no pair in a mini-batch satisfied the margin, L_{rank} was set to zero.

The total mini-batch objective was:

$$L = \frac{1}{B} \sum_{i=1}^B (L_{micro}^{(i)} + \alpha_e L_{TE}^{(i)}) + \beta L_{rank}, \beta = 0.2$$

To stabilize optimization, α increased linearly from 0.2 to 4.0 over the first 30% epochs and remained at 4.0 thereafter. Validation loss used $\alpha = 4.0$ throughout. CDS boundaries and reading frames, when available, were used only to construct training losses. TRACE inference required only RNA sequence, species and the cellular context expression vector.

During training, species labels and cellular context inputs were independently masked with probability 0.1. Masked species labels were assigned to an unknown category, whereas masked cellular contexts received a zero expression vector. When only the species label was masked, the expression vector was replaced by the cross-species mean for that cell type when available, and otherwise by zero. Remained 30% of expression vectors were continuously interpolated toward zero during training. Gaussian noise (s.d. = 0.1) was further added to expression vectors during training. Masking, interpolation and noise were disabled at inference.

Training used AdamW ($\beta_1 = 0.9, \beta_2 = 0.9, \epsilon = 1 \times 10^{-9}$ and peak learning rate = 1×10^{-3}) for 60 epochs, with early stopping disabled. Weight decay was 0.01, except for biases, layer-normalization parameters and embeddings, for which it was zero. The learning rate increased linearly during the first 30% of epochs and then followed cosine decay. Gradients were clipped at a norm of 2.0.

A distributed length-bucket sampler grouped similarly sized transcripts. Species–cellular-context classes with at least 6,000 profiles were randomly capped at 6,000 profiles per epoch; smaller

classes were retained without upsampling. Training used bfloat16 automatic mixed precision, 50 profiles per GPU and no gradient accumulation. The effective batch size was 150 across three NVIDIA H20 GPUs (96 GB each) using DistributedDataParallel.

Ablation experiments

We used matched-input ablations to distinguish the contributions of cellular-context conditioning and sequence modeling. LN-model was a sequence-only pre-LayerNorm Transformer that retained global self-attention but removed adaptive cellular-context conditioning. Conv-model replaced the Transformer encoder with parameter-matched residual one-dimensional convolutional blocks, providing a local-sequence baseline. All variants used the same nucleotide input, prediction head, training objectives and chromosome-disjoint evaluation framework. Comparisons among TRACE-Augment, TRACE-Real, TRACE-Zero and LN-model distinguished measured cellular-context information from the adaptive-normalization parameterization, whereas the LN-model–Conv-model comparison tested global self-attention against local convolution.

Cellular-context conditioning was further evaluated with three strategies: Zero, in which the expression vector was set to zero; Real, in which the measured expression vector was supplied; and Augment, in which expression vectors were stochastically masked or continuously interpolated toward zero during training. To assess how training-set diversity affected generalization, each strategy was trained using 5, 22 or 40 cellular environments and evaluated on the same chromosome-held-out transcripts from 26 unseen human cellular environments. Results for the five-context model ablation are summarized in Supplementary Table 2. Full model configurations, augmentation parameters and evaluation procedures are provided in Supplementary Methods.

Benchmarking on translation tasks

Translation efficiency (TE) evaluation. We compiled polysome profiling, protein-to-RNA ratio and SILAC pulse-labeling datasets. For TRACE, the TE indicator was the mean predicted occupancy over frame-0 positions in the annotated CDS. RiboNN and RiboDecode directly output transcript-level TE values. Sequence-derived baselines included 5' UTR length, Kozak

strength and upstream AUG count. We computed Spearman ρ between predictions and experimental measurements across test transcripts.

Translational dynamics evaluation. We used nucleotide-resolution P-site density profiles from held-out Ribo-seq data. For each transcript, Spearman ρ was computed between predicted and observed profiles across nucleotide positions. TRACE and Translatomer were evaluated on full-length outputs. Riboformer and RiboMIMO were retrained on CDS-restricted codon features because pretrained multi-context eukaryotic parameters were unavailable. Reference baselines compared profiles across runs within a study and across studies of the same cell type.

Riboformer sequence-only retraining. Published Riboformer combines aligned 40-codon sequences with matched-control ribosome-density inputs. For the sequence-only comparison, we removed the density branch and its fusion with the sequence representation. The sequence encoder was connected directly to the ribosome-density prediction head. Ribo-seq-derived P-site densities served only as training and evaluation targets. We used the same human chromosome partitions as for TRACE.

De novo ORF identification. We used public mass spectrometry data to define translated-ORF evidence. TRACE performed annotation-free inference to generate nucleotide-resolution ribosome occupancy profiles. It then scanned these profiles for candidate ORFs and scored periodicity, occupancy uniformity, mean occupancy, start-site step-up and stop-site drop-off. RibORF, Ribo-TISH and RiboTIE received RNA sequence and Ribo-seq data, whereas TranslationAI received RNA sequence alone. Sequence-derived baselines ranked candidates by ORF length or host-transcript expression.

Model explainability and attention analysis

mRNA sequence optimization algorithm

Experimental validation of optimized mRNA designs

Tumor neoantigen candidate prioritization from transcriptome data

Paired-end RNA-seq from tumor and adjacent-normal tissue was processed through a common assembly and quantification pipeline. Reads were trimmed with fastp and aligned to GRCh38 using STAR in two-pass mode. StringTie assembled each sample, and StringTie --merge produced a unified transcript annotation. featureCounts generated transcript abundance and splice-junction counts.

Tumor-specific transcripts were identified using expression-level and splice-junction evidence. The expression track required tumor TPM ≥ 1.0 , log2 fold change ≥ 2.0 and maximum normal TPM < 0.5 . The junction track required mean tumor CPM ≥ 2.0 , log2 fold change ≥ 2.0 and maximum normal CPM < 1.0 . Candidates were filtered against GTEx v11 using both summarized expression and direct re-quantification in raw GTEx alignments.

TRACE inference on verified tumor-specific transcripts was annotation-free. The tumor gene-expression profile supplied the cellular context. TRACE predicted a nucleotide-resolution ribosome occupancy profile for each transcript. Candidate ORFs were scored using periodicity, occupancy uniformity and start-site and stop-site transitions. ORF-level expression was estimated from transcript TPM and mean predicted occupancy. Candidate nomination did not require matched tumor Ribo-seq or mass spectrometry data.

Peptide candidates were filtered against matched-normal TRACE predictions and the canonical human proteome. Eight- to eleven-amino-acid peptides were evaluated with netMHCpan 4.1 for the patient HLA-A alleles. Weak and strong binders were retained and prioritized by estimated ORF expression and predicted presentation probability.

Functional evaluation of predicted neoantigen candidates

Data availability

The curated Ribo-seq and RNA-seq dataset, processed ribosome occupancy profiles, and model predictions are available at [repository URL]. Public datasets used in this study are listed in Supplementary Table 1.

Code availability

Model code and training scripts are available at: [<https://github.com/Chunfu-Shawn/TRACE>].
Processed datasets (for training, validation and test) and pretrained checkpoints are archived on
Zenodo: [DOI:10.5281/zenodo.xxxxxxx]

References

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Author contributions

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Competing interests

[To be completed.]

Figure legends

Fig. 1 | Dataset overview. (a) CDS ribosome occupancy in canonical protein-coding transcripts.
(b) Cellular context-associated occupancy patterns.

Fig. 2 | TRACE model and predictions. (a) Architecture. (b) Predicted and observed nucleotide-
resolution ribosome occupancy profiles on held-out transcripts. (c) CDS enrichment. (d)
Periodicity by transcript biotype.

Fig. 3 | Benchmarking a unified translation representation. (a) Three benchmark tasks. (b,c)
Translation-dynamics performance by transcript length and occupancy. (d) Performance across
cellular contexts, including held-out SW480. (e–g) Translation-efficiency comparisons with
polysome profiling, protein-to-RNA ratios and SILAC.

Fig. 4 | Translation-associated patterns in TRACE predictions. (a) Transcript-region lengths and
predicted TE. (b) RNA minimum free energy. (c) Aggregated attention. (d) Kozak-associated

507 attention. (e) Frame-resolved attention. (f) In silico codon substitutions. (g,h) Predicted effects of
508 upstream start codons.

509 Fig. 5 | Forward mRNA sequence design in cultured cells. (a) Graded Fluc and mCherry designs.
510 (b) Gluc comparison with wild-type, RiboDecode and LinearDesign. (c) Cellular-context-
511 specific designs in HEK293T, HeLa and HepG2 cells.

512 Fig. 6 | Reverse discovery of tumor neoantigen candidates. (a) Candidate-prioritization pipeline.
513 (b) IFN- γ ELISpot. (c) Patient-derived organoid co-culture assay.